1 Quasiparticle-interference results for a two-dimensional superconducting multiorbital Weyl SSH model

The emergence of topological superconductivity in a two-dimensional (2D) Weyl system, composed of stacked Su-Schrieffer-Heeger (SSH) chains was studied by Rosenberg and Manousakis [CITE: Rosenberg, Aryal PRB 100 104522 (2019)]. They showed that an attractive Hubbard interaction between spinful electrons leads to a topological trivial supercondictor. Next, a pairing interaction that couples spinless fermions on opposite sublattices within the same unit cell was shown [CITE: Rosenberg and Manousakis arXiv:2203.12004v1] to induce a topological superconducting state, characterised by a gap function with a non-trivial phase, as well as Majorana and Fermi arc edge states. We reproduce the results in a finite model with open-boundary conditions along the direction of SSH dimerisation and periodic boundary condition on the perpendicular axis. We then investigate the model with an impurity, self-consistently, in the mean-field limit. We show the expected quasiparticle-interference, superconducting gap spectrum and local density of states in such a system. Later, we also wish to investigate $v = w = t_d$, for the purposes of spontaneous Pierls or Jahn-Teller distortion (what is this? CDW?), as well as a system with periodic-boundary conditions and a wall of impurities.

1.1 Reproducing results

The Hamiltonian is given by

$$\begin{aligned}\hat{\mathcal{H}}_0 &= -\sum_{1\leq x,y\leq N}\left[v\hat{c}^{(A)\dagger}_{x,y}\hat{c}^{(B)}_{x,y} + w\hat{c}^{(A)\dagger}_{x,y}\hat{c}^{(B)}_{x-1,y} + \text{h.c.}\right] \\ &\quad -\sum_{1\leq x,y\leq N}\left[t_d\hat{c}^{(B)\dagger}_{x,y}\hat{c}^{(A)}_{x,y\pm1} + t_d\hat{c}^{(B)\dagger}_{x,y}\hat{c}^{(A)}_{x+1,y\pm1} + \text{h.c.}\right] \\ &\quad -\sum_{1\leq x,y\leq N}\left[\left(\mu + s(-1)^m\right)\hat{n}^{(m)\dagger}_{x,y}\right], \end{aligned} \tag{1}$$

where the fermion operator $\hat{c}^{(m)\dagger}_{x,y}$ creates an operator on atom $m \in \{A, B\}$ within the unit cell located at $\mathbf{R}_{x,y} = x\mathbf{b}_0\hat{x} + y\mathbf{b}_1\hat{y}$, where $\mathbf{b}_0, \mathbf{b}_1$ are the lattice basis vectors, $\hat{n}^{(\alpha)\dagger}_{x,y}$ is the number operator, μ is the chemical potential and s is a rigid shift of the atomic potentials. The intra- and inter-unit-cell hoppings strengths in the $\hat{x}$ direction are given by v and w, which form the terms of the standard SSH model. These 1D SSH chains are coupled by a diagonal hopping t_d, which form a two-dimensional stacking. The lattice is depicted with $N = 3$ unit cells in Figure 1. Following [CITE: Rosenberg and Manousakis arXiv:2203.12004v1], without loss of generality, we set $v = 0.6, w = 1.2$, the topological phase of the SSH model, and $t_d = 0.9$.

The solution to the Hamiltonian is given by the Green's function, which we remind the

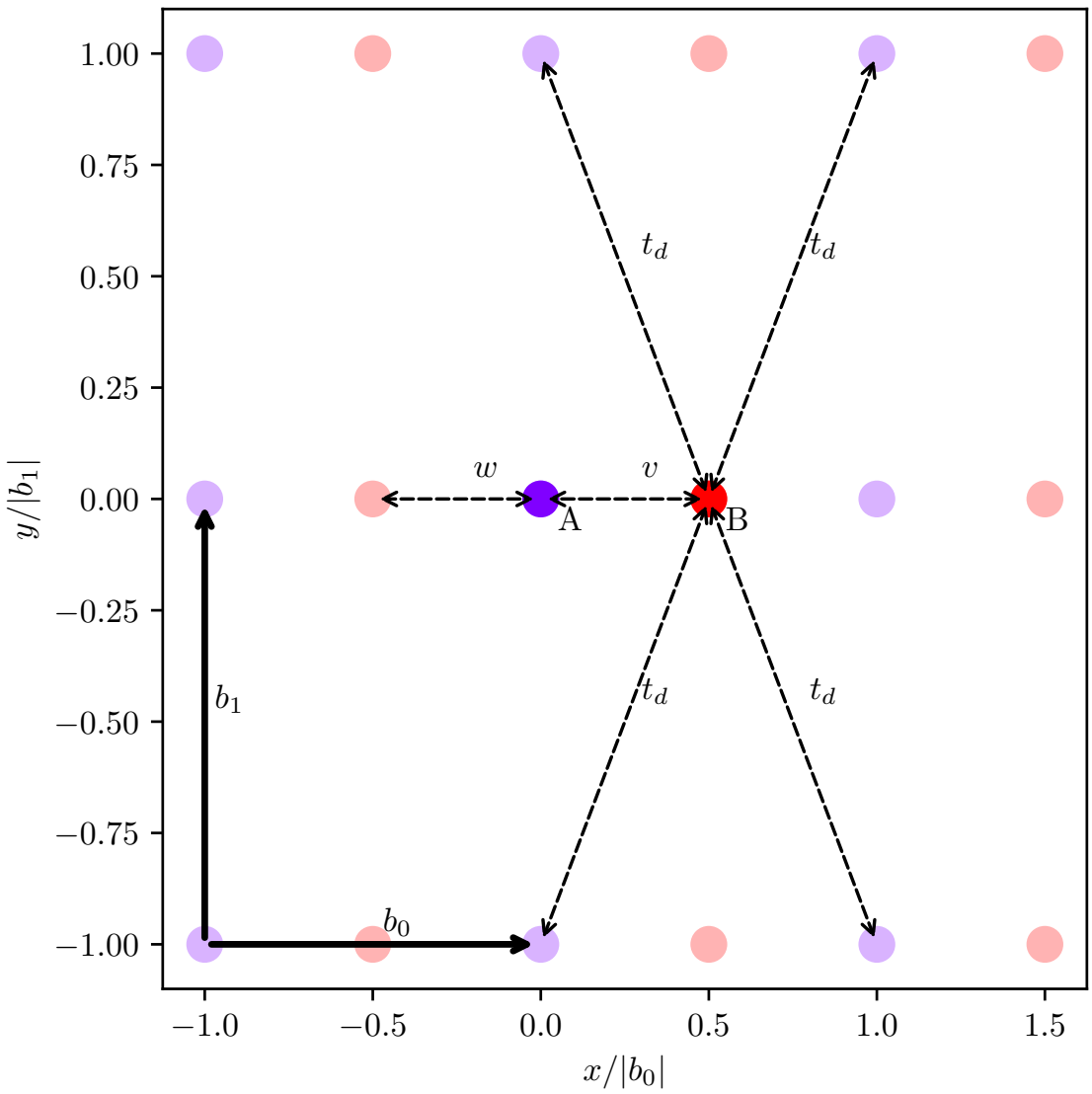
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
$y/|b_1|$
−1.0
−0.5
0.0
0.5
1.0
1.5
$x/|b_0|$
t_d
t_d
w
v
A
B
b_1
t_d
t_d
b_0

Figure 1:

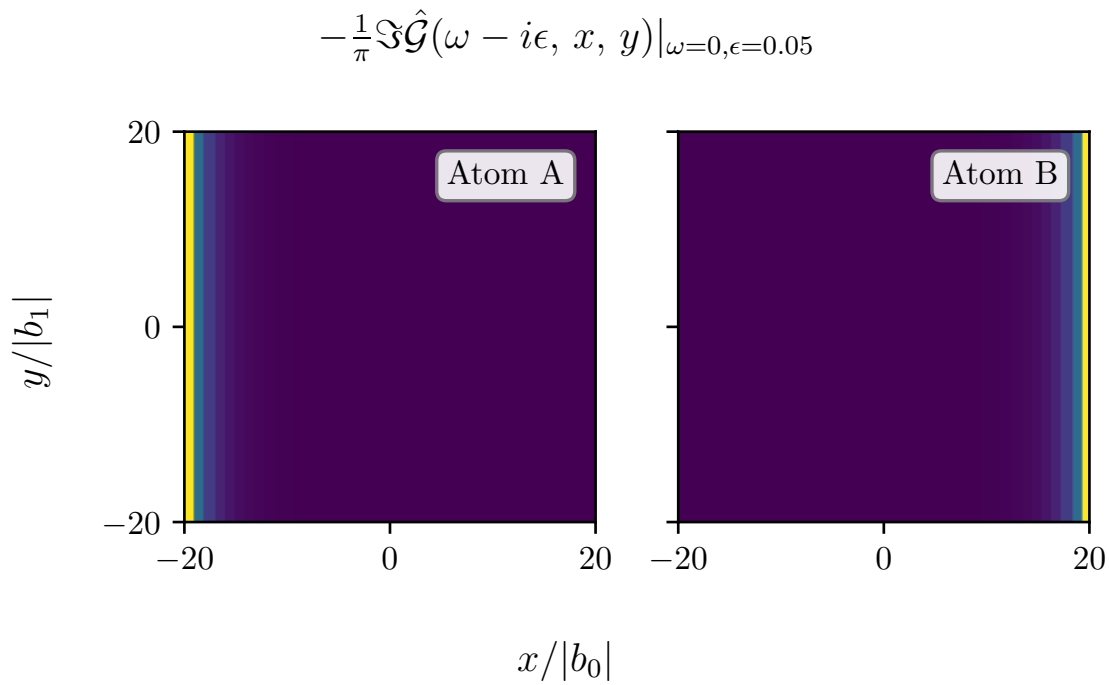

Figure 2:

reader, is given by

$$\hat{\mathcal{G}}(\omega - i\epsilon, x, y) \equiv \sum_n \left[\frac{\psi_n(x,y)\psi_n(x,y)^\dagger}{\omega - i\epsilon - E_n} \right], \tag{2}$$

where $\psi_n(x, y)$ is the eigenvector associated with the nth eigenenergy, resolved at the point on the lattice (x, y). The Green's function matrix has a component for each atom. The Green's function is experimental observables through the *local density of states*,

$$\text{LDOS}(\omega; x, y) \equiv -\frac{1}{\pi}\Im\hat{\mathcal{G}}(\omega - i\epsilon, x, y), \tag{3}$$

which is a measure of the density of states at a given point, at a specified applied bias energy ω and resolution ϵ. The local density of states for each atomic site with the unit cells are given in Figure 2, showing the existence of zero energy edge states.

Resolving the local density states at the level of the unit cell (summing over the atoms) and integrating over the y-axis, we study the *spectrum* of the system. The spectrum shows edge modes present at zero energy at the edges, which are not present in the bulk. These are distinct signatures of Majorana modes. In addition to localised density at the edge, there exists a 'tail' of density which the authors investigate further.

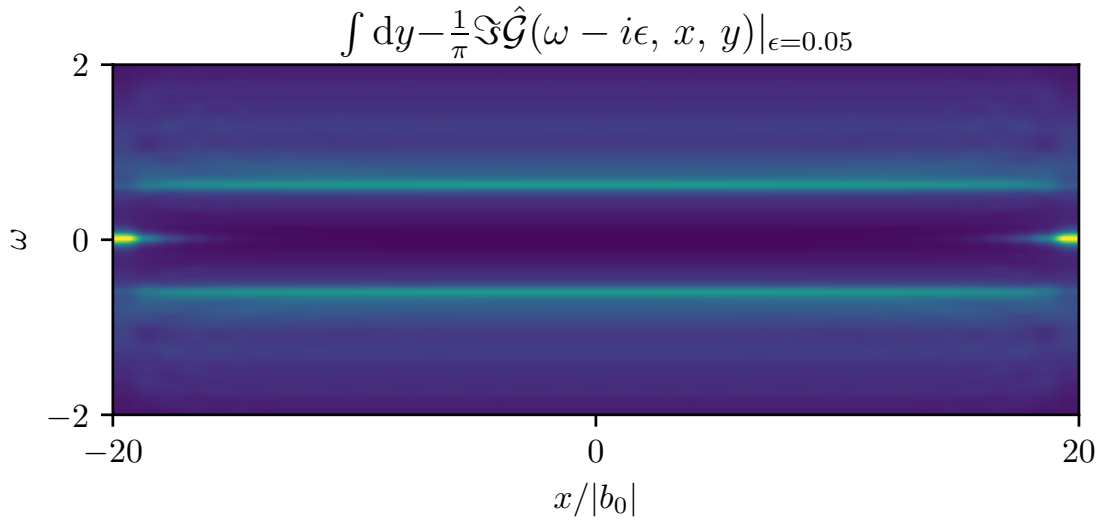

Figure 3:

Majorana modes exist pinned to *Dirac points*, which are the cross point of two bands. Studying the band structure in Figure 10, we observe a zero-mode between the Dirac points, which is known as a *Bogoliubov-Fermi arc*.

In order to determine whether the modes are topological modes, it must be shown that they their wavefunctions are distributed nonlocally across the system. We plot the wavefunction via the related *spectral density* of the Green's function at the point k_y at which the Bogoliubov-Fermi arc and Majorana modes are are located, Figure 12. As a function of x-position, we observe that indeed, the modes are spread non-locally and that the arc tails.

Mention Bo Huang's thesis results on the influence of the shape of the lattice on the topological modes.

1.2 Interactions and mean-field solution

We add an interaction term to model multiorbital superconductivity within the unit-cell, U_v, and an interorbital intercell Coulomb repulsion term, U_w. The interacting contribution to the Hamiltonian is

$$\hat{\mathcal{H}}_I = -U_v \hat{c}^{(A)\dagger}_{x,y}\hat{c}^{(B)\dagger}_{x,y}\hat{c}^{(A)}_{x,y}\hat{c}^{(B)}_{x,y} + U_w \hat{c}^{(A)\dagger}_{x,y}\hat{c}^{(B)\dagger}_{x-1,y}\hat{c}^{(B)}_{x-1,y}\hat{c}^{(A)}_{x,y}. \tag{4}$$

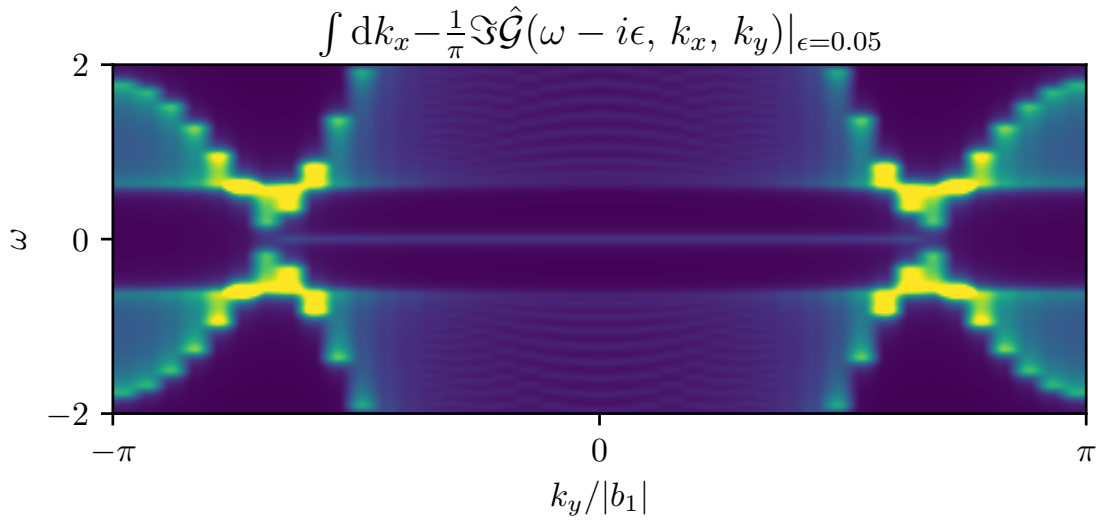

Figure 4:

The mean-field Hamiltonian is

$$\hat{\mathcal{H}}_I' = \sum_{x,y}\left[\sum_m \left[\phi_m \hat{n}_{x,y}^{(m)}\right] + \Phi_v \hat{c}_{x,y}^{(A)}\hat{c}_{x,y}^{(B)} + \Phi_w \hat{c}_{x,y}^{(A)}\hat{c}_{x-1,y}^{(B)} \right.$$
$$\left. + \Delta_v \hat{c}_{x,y}^{(A)\dagger}\hat{c}_{x,y}^{(B)\dagger} + \Delta_w \hat{c}_{x,y}^{(A)\dagger}\hat{c}_{x-1,y}^{(B)\dagger} + \Delta_v^* \hat{c}_{x,y}^{(A)}\hat{c}_{x,y}^{(B)} + \Delta_w^* \hat{c}_{x,y}^{(A)}\hat{c}_{x-1,y}^{(B)}\right], \quad (5)$$

where ϕ_v is the *Hartree* term which renormalises the chemical potential, allowing for a Peirl's instability. Φ_v is a *Fock* term, describing renormalistion of the hopping parameter v. The additional *Gorkov* terms Δ_v capture fermion-fermion pairing, relevant to theories of magnetism and superconductivity. These terms are known as the method of *(superconducting) density functional theory, SC-DFT*. The self-consistent solution to the mean field

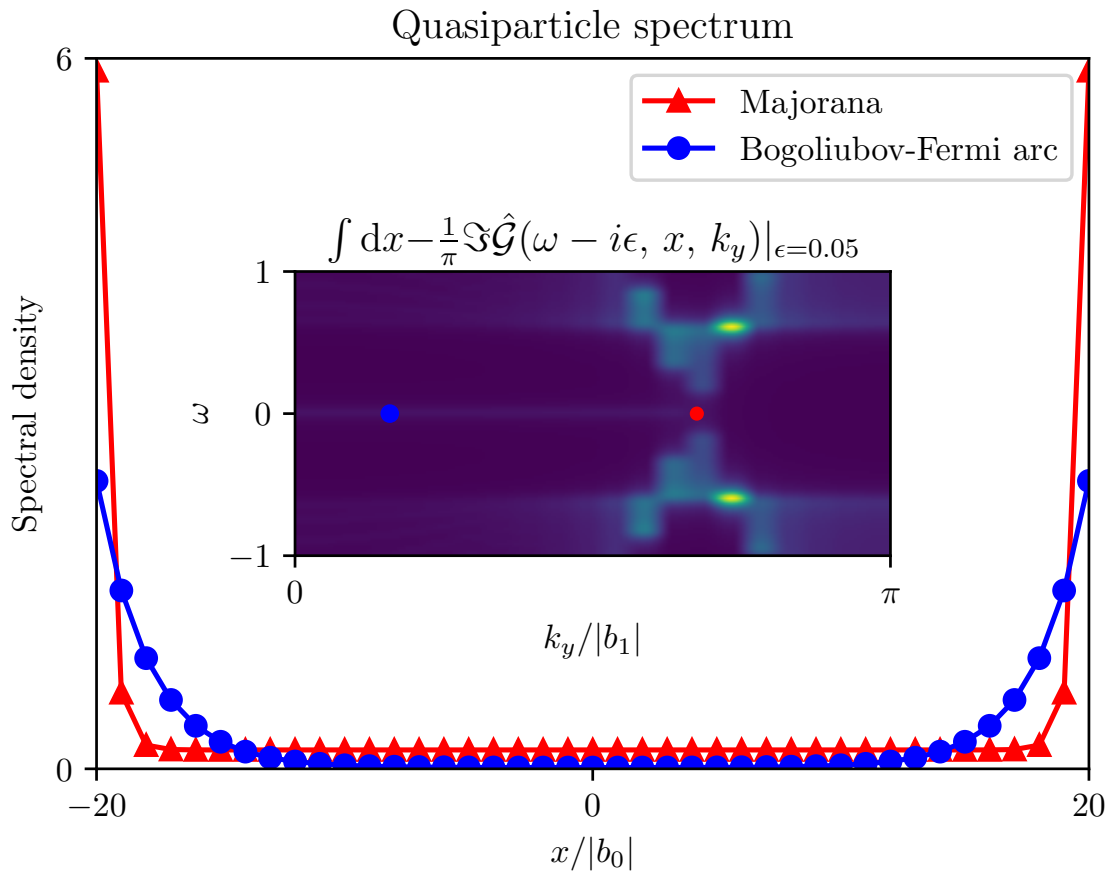
Quasiparticle spectrum
Majorana
Bogoliubov-Fermi arc
$\int \mathrm{d}x - \frac{1}{\pi}\Im\hat{\mathcal{G}}(\omega - i\epsilon,\, x,\, k_y)|_{\epsilon=0.05}$
ω
$k_y/|b_1|$
Spectral density
$x/|b_0|$
6
0
−20
0
20
1
0
−1
0
π

Figure 5:

is the saddle point $\langle \hat{\mathcal{H}}_I \rangle = \langle \hat{\mathcal{H}}_I' \rangle$. The solutions are

$$\rho_m = (-U_v + U_w))\langle \hat{n}_{x,y}^{(m')} \rangle, \tag{6}$$

$$\Phi_v = -U_v \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x,y}^{(B)} \rangle, \tag{7}$$

$$\Phi_w = U_w \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x-1,y}^{(B)} \rangle, \tag{8}$$

$$\Delta_v = +U_v \langle \hat{c}_{x,y}^{(A)} \hat{c}_{x,y}^{(B)} \rangle, \tag{9}$$

$$\Delta_w = -U_w \langle \hat{c}_{x,y}^{(A)} \hat{c}_{x-1,y}^{(B)} \rangle, \tag{10}$$

$$\Delta_v^* = +U_v \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x,y}^{(B)\dagger} \rangle, \tag{11}$$

$$\Delta_w^* = -U_w \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x-1,y}^{(B)\dagger} \rangle, \tag{12}$$

where m' refers to the opposite atom to m. Note the sign of the Gorkov terms due Fermionic anticommutation.

For simplicity and because we are in particular interested in the multiorbital pairing due to U_v, we neglect the intercell interaction U_w.

1.3 Results with impurities

The impurity is located at the centre with coupling strength $V = 1.21$.

The x-axis is glued to periodic boundary conditions.

Our investigation shows no interesting interplay between impurities and topological modes. However, impurities can be used to generate quasiparticle-interference patterns as an experimental diagnosis of topological modes.

1.4 Quasiparticle-interference patterns for Majorana and Bogoliubov-Fermi arcs

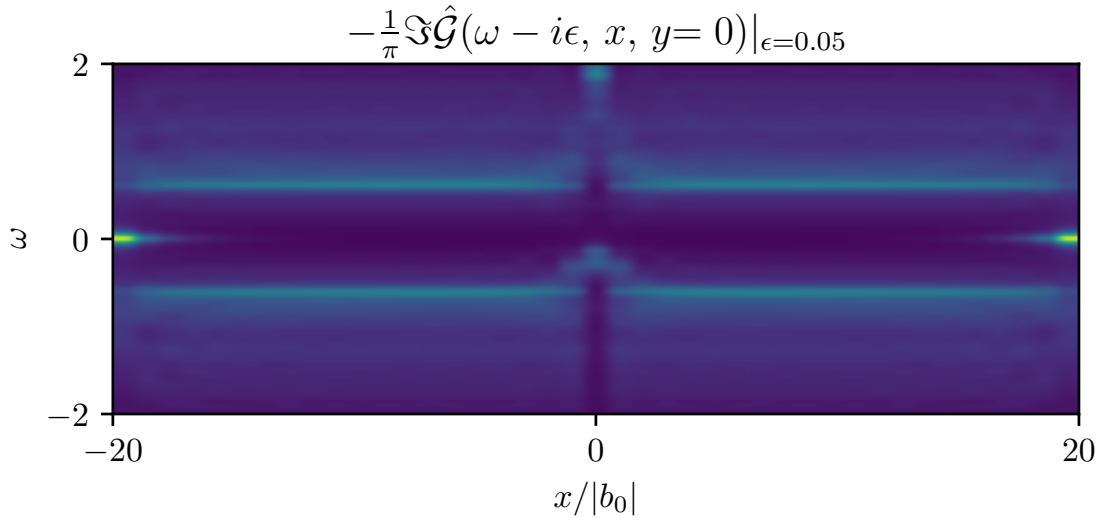

Figure 6: Local density of states integrated over the intracell atomic sites. The chemical potential is $\mu/t = 0.00$ and the lattice is 41×41 with periodic boundary conditions along the vertical, and open boundary conditions along the horizontal. The impurity has coupling strength $V/t = 1.21$.

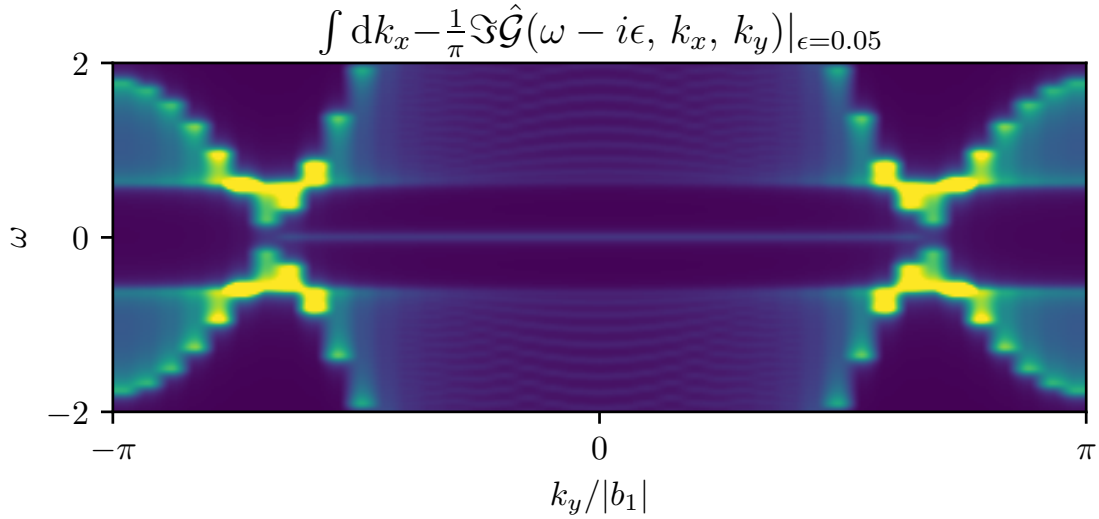

Figure 7: Band structure integrated over the axis of the SSH dimers, k_x, with open boundary conditions along the x axis. An additional mode crosses zero energy at $k_y = \pm\pi$.

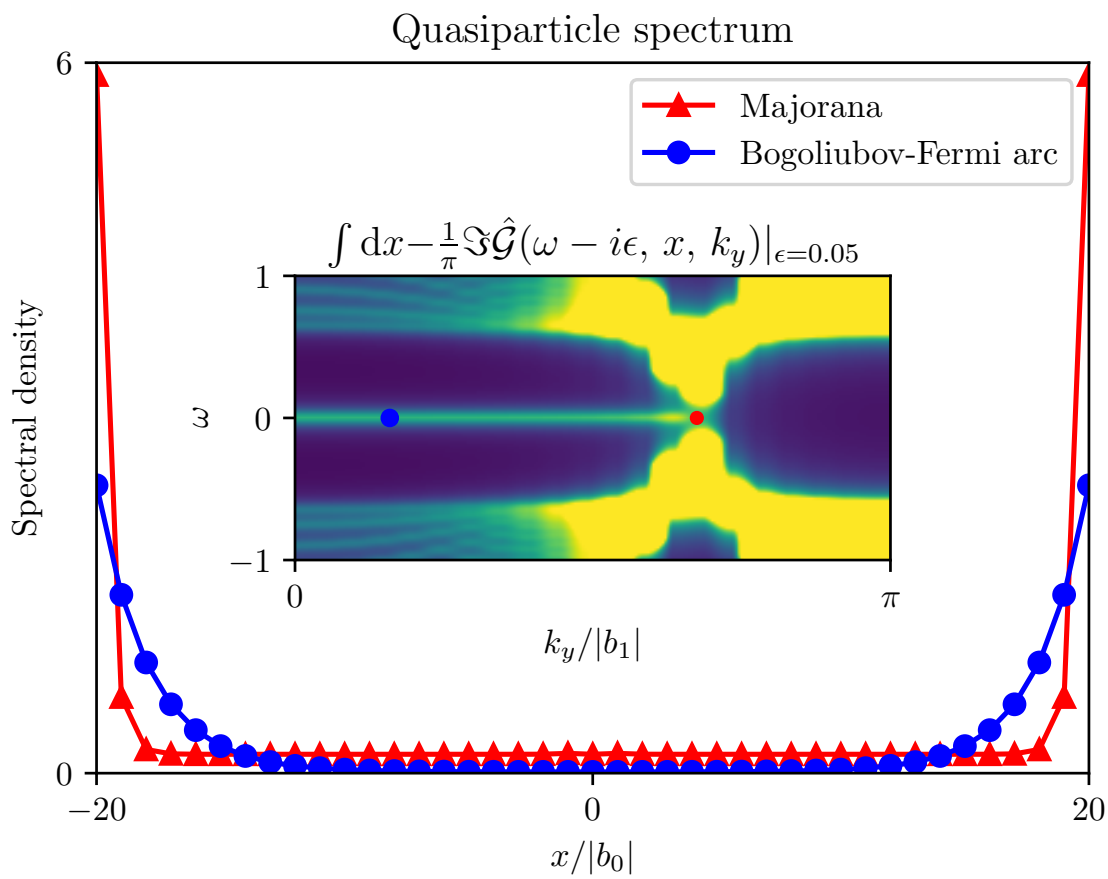

Figure 8: Topological modes in the presence of an impurity. No additional topological effects seen.

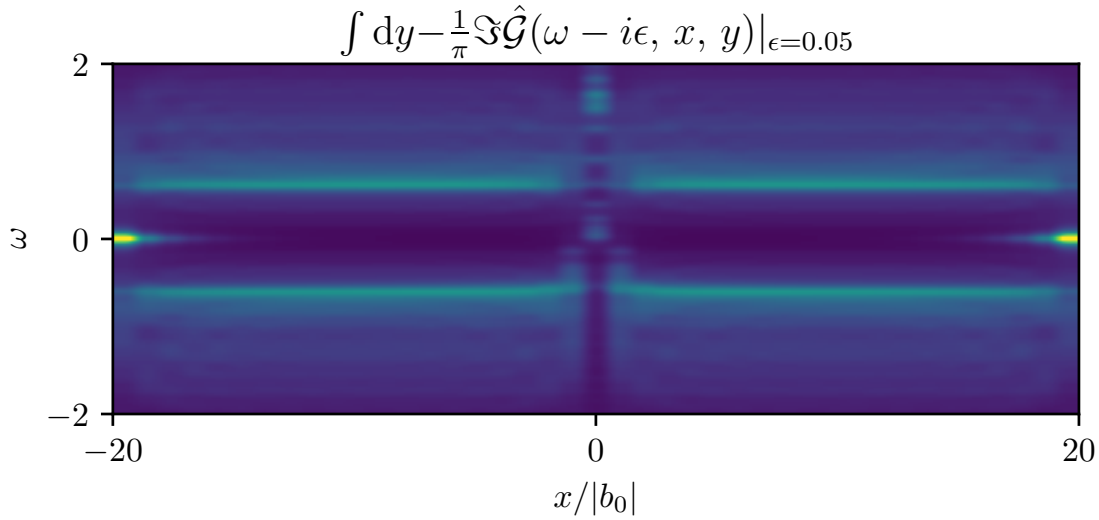

Figure 9: Local density of states with an impurity at the centre. The chemical potential is $\mu/t = 0.00$ and the lattice is 41×41 with periodic boundary conditions along the vertical, and open boundary conditions along the horizontal. The impurity has coupling strength $V/t = 1.21$. No additional topological features are seen.

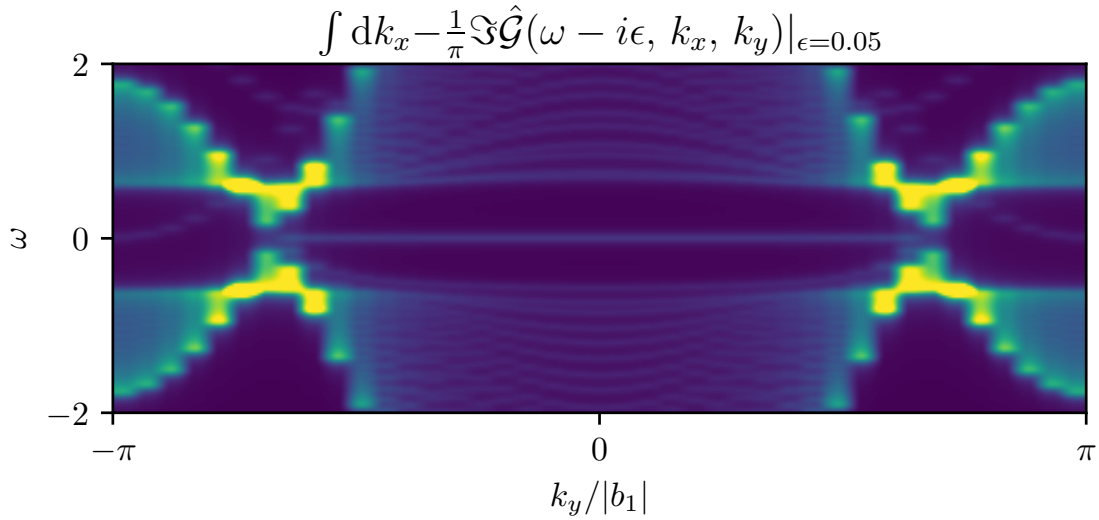

Figure 10: Local density of states integrated over the intracell atomic sites. The chemical potential is $\mu/t = 0.00$ and the lattice is 41×41 with periodic boundary conditions along the vertical, and open boundary conditions along the horizontal. The impurity has coupling strength $V/t = 1.21$.

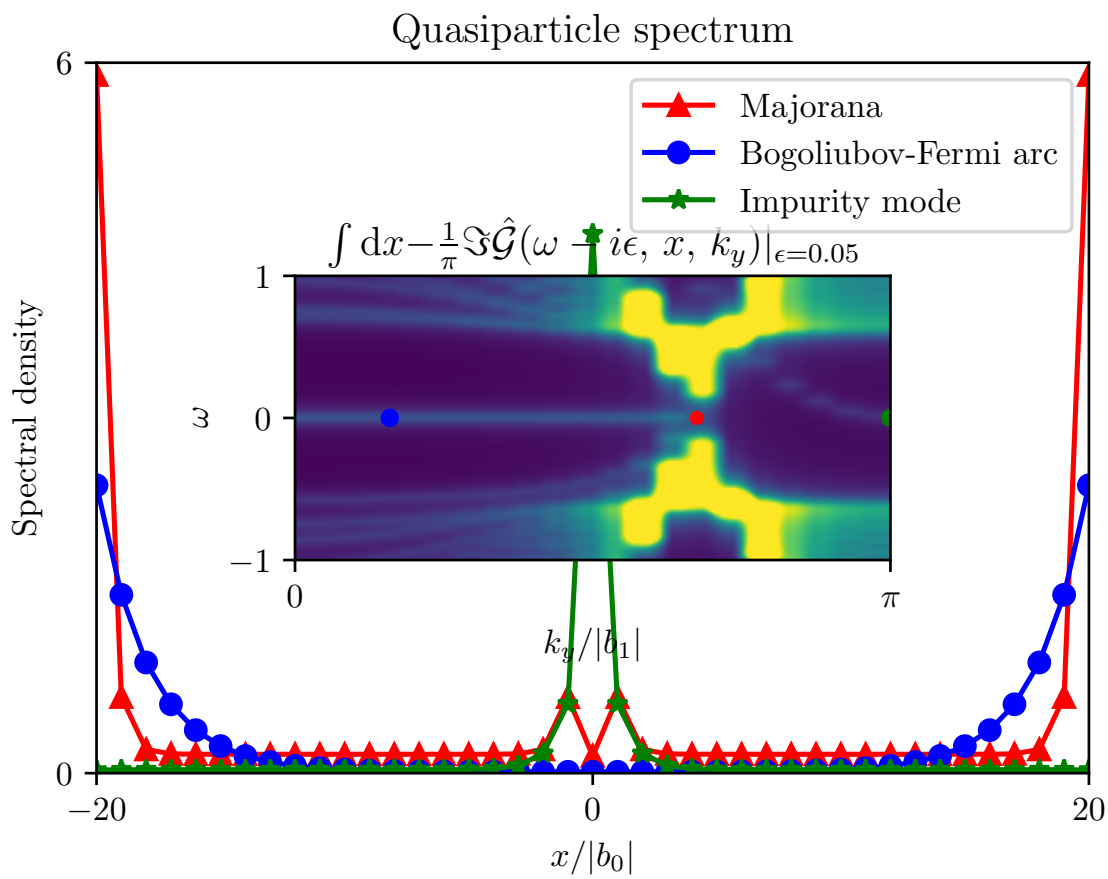

Figure 11: Topological modes and an additional mode $k_y = \pi$. The nature of the additional mode is to be investigated.

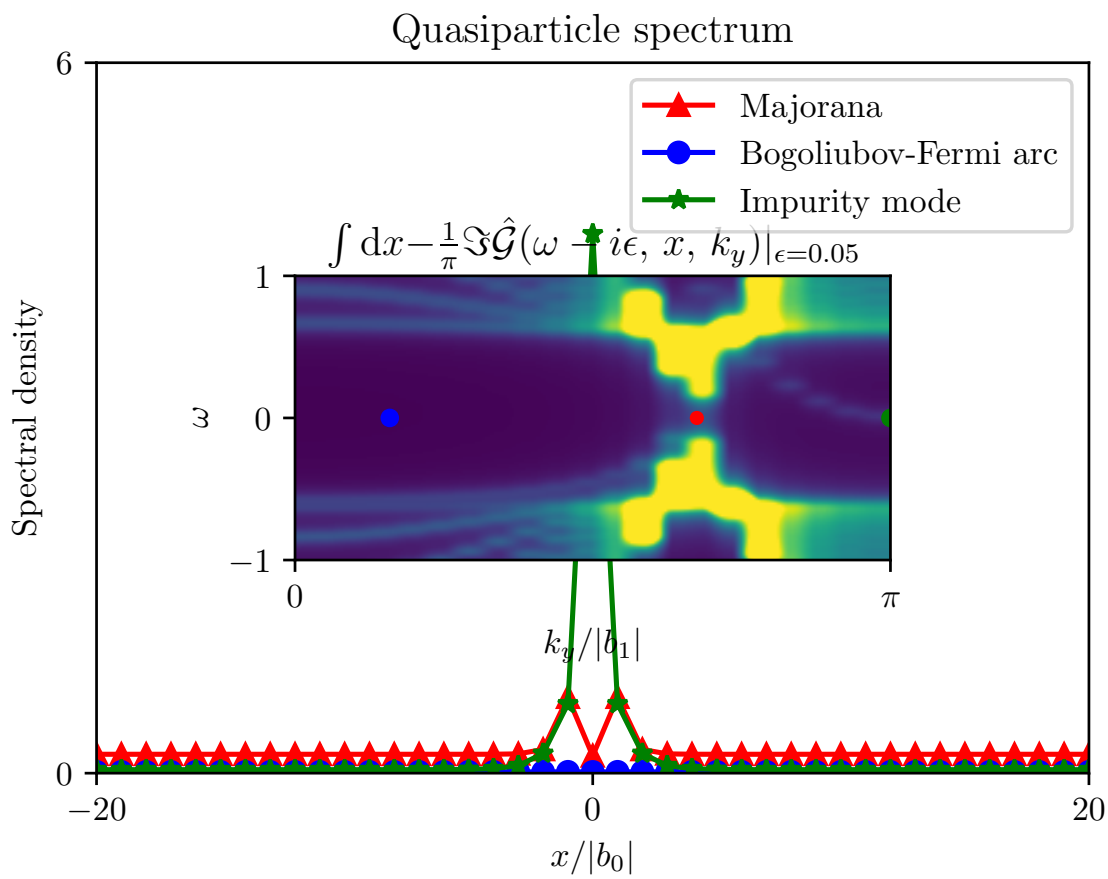

Figure 12: Investigation of the modes with periodic boundary conditions along both axes. The additional mode is shown to be due to the presence of a wall of impurities.

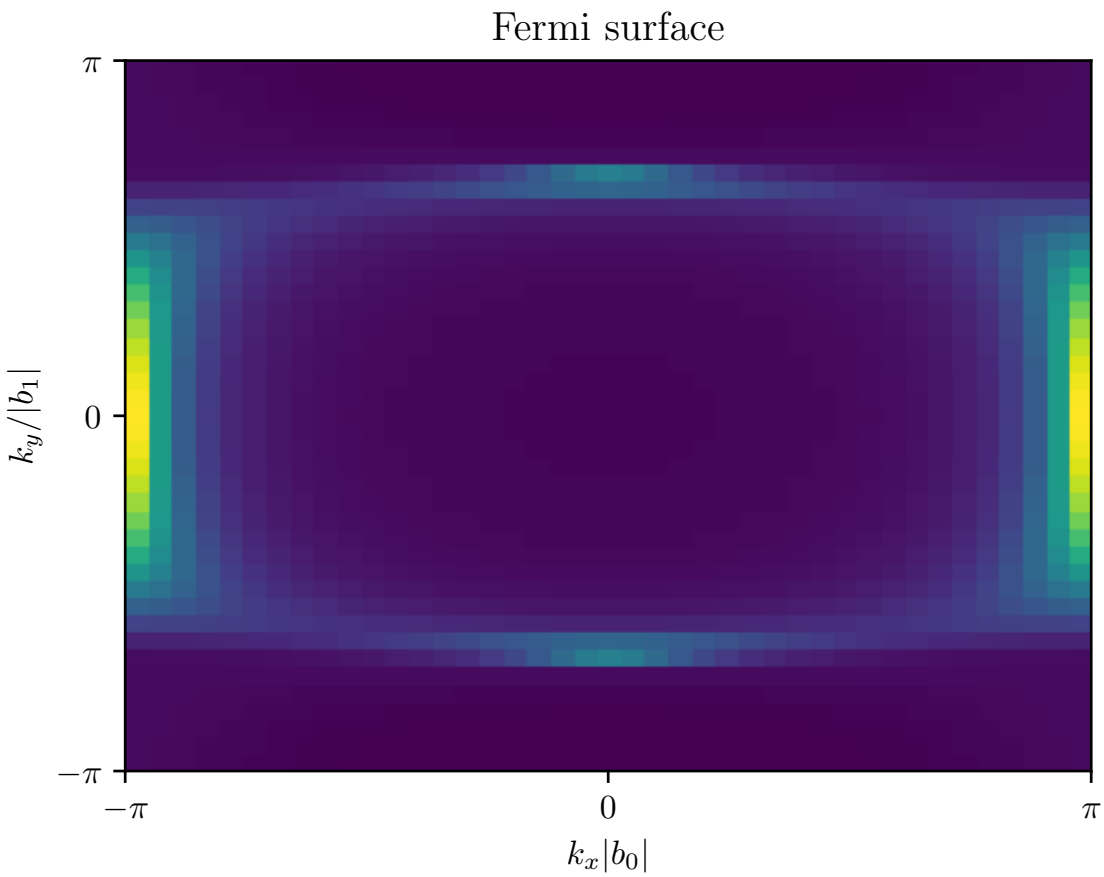

Figure 13: Fermi surface with a central impurity with coupling strength $V = 1.21$ and chemical potential $\mu = 0.0$, in the normal state. The hopping parameters are $t_d = 0.9, v = 0.6$ and $w = 1.2$. Notice density concentrated near $k_y \approx \pi/2$ corresponding to the Majorana mode, and density around $k_x \approx \pm\pi$ corresponding to the Fermi arc.